

Octavio Villalobos

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EDUCATION

University of California, Santa Cruz

Master of Science in Computer Science — GPA: 3.74/4.00

June 2026

Santa Cruz, CA

University of California, Santa Cruz

Bachelor of Science in Computer Science, With Honors — GPA: 3.60/4.00

June 2024

Santa Cruz, CA

SKILLS

Programming Languages: Python, C/C++, Java, JavaScript, TypeScript, C#, SQL, HTML/CSS, Angular

Front-end Development: React.js, React Native (Expo), Tailwind CSS, HTML5

Backend & Databases: FastAPI, Express.js, Node.js, RESTful APIs, PostgreSQL, Supabase

AI/ML & Generative AI: PyTorch, TensorFlow, Scikit-learn, LangChain, LangGraph, Hugging Face, OpenAI API, Claude API, LoRA, QLoRA, Ollama, ChromaDB, MLflow, Weights & Biases (W&B), Pandas, Matplotlib, Seaborn

Cloud & Infrastructure: Google Cloud Platform (GCP), AWS, Docker, Linux

DevOps & Testing: Git, GitHub Actions, CI/CD Pipelines, Pytest

EXPERIENCE

Research Project Lead @ Sloth Lab – UCSC

Oct 2025 – Present

Santa Cruz, CA

- Led a 3-person team extending the Lyra opinion dynamics framework into a scalable multi-agent simulation system for personality-driven NPC interactions and emergent social behavior
- Designed agent architectures with 23-dimensional state tracking and **prospect-theory-based** persuasion modeling, implementing per-agent **Actor-Critic (A2C)** RL policies across 5 personality archetypes
- Developed clustering and behavioral-analysis pipelines using **MLflow** experiment tracking, validating emergent interaction patterns across 1,000+ large-scale simulation runs
- Co-authored a research paper submitted to **AIIDE 2026** (under review) exploring emergent social behavior, conversational dynamics, reinforcement learning, and multi-agent interaction systems

AI Engineer Intern @ Silicon Valley Certification Hub

June 2025 – Present

Palo Alto, CA

- Developed and deployed an **LLM-powered** coaching platform (**Flask**, **AWS**) built around a RAG-based retrieval pipeline that generates executive-style scenario assessments and coaching feedback
- Engineered rubric-based scoring pipelines using structured **prompt engineering** and configurable grading curves to evaluate answer relevance, ethical risk coverage, and actionability
- Fine-tuned **Gemma** and **LLaMA** models using **LoRA**-based parameter-efficient techniques, improving factual alignment by 42.3%, response consistency by 25.1%, and contextual relevance 33.1% for certification coaching
- Implemented role-conditioned scenario generation and optional **ElevenLabs** text-to-speech integration, supporting realistic AI interview simulations across 12 industries

Teaching Assistant @ University of California, Santa Cruz

Sep 2024 – June 2026

Santa Cruz, CA

- Delivered lectures on databases, AI, LLMs, and technical writing to 150+ students across 4 UCSC courses
- Led weekly discussion sections for 30 students, developing instructional materials and hands-on lesson plans
- Designed coursework, assignments, and exams across 6 courses to improve alignment between lectures

PROJECTS

FormulaForecast | Python, Pandas, NumPy, PyTorch, scikit-learn

Nov 2025 – Jan 2026

- Developed a **deep learning** framework in **PyTorch** for predicting Formula 1 top-10 race finishing positions, leveraging historical race telemetry and driver-performance data from the **FastF1 API**
- Engineered 9 domain-specific features, implementing **StandardScaler** normalization, **3-sigma outlier clipping**, and filtering pipelines to remove DNFs and high-variance race incidents
- Designed and trained a 3-layer **MLP neural network (128→64→32)** using **ReLU** activations, **Dropout**, **Batch Normalization**, and **He/Kaiming initialization**

- Achieved an MAE of 1.585 and 88.7% within-3 prediction accuracy on unseen 2025 race data while reducing prediction error by 41% compared to baseline full-grid neural network models

Insight Deck | *Python, React, Flask, LangChain, Vector DB, ChromaDB, RAG*

Apr 2025 – Dec 2025

- Co-developed a time-aware hybrid retrieval pipeline combining **BM25**, **ChromaDB** vector search, and metadata-aware reranking, improving form-type retrieval accuracy across multi-year financial corpora
- Orchestrated a multi-agent **RAG** workflow using **LangChain**, coordinating Search, Parser, Critique, and Slide Generation agents to automate investor presentation creation from scraped financial and company data
- Engineered **dynamic query-generation** and **prompt-planning** pipelines with overlapping Markdown chunking, semantic retrieval, and contextual refinement workflows, improving content population from 74.2% to 98% while reducing slide-planning latency by 49%
- Reduced slide-generation failures and low-content outputs by engineering an iterative **LLM critique** and repair pipeline from 70% to 90% and improving presentation completeness by 10–15% per iteration

LLMs are ACDs | *Python, CybORG, CAGE 4, OpenAI API, HuggingFace, Ollama*

Apr 2025 – Dec 2025

- Co-authored an **IEEE CAI 2025** paper developing the first framework for **LLM** and **RL** agents in **CybORG CAGE 4** multi-agent cybersecurity environment, enabling interpretable autonomous cyber defense simulations
- Collaborated within a 6-person research team to create an 8-bit communication protocol enabling coordination across agents (**LLM + RL**), supporting real time threat sharing and distributed defense across 4 network zones
- Benchmarked **OpenAI**, **Ollama**, and **Deepseek** against **RL** baselines across 5 attacking agents, applying behavioral analysis and reward-based evaluation to measure interpretability, action quality, and defensive performance
- Built **Python** adapters integrating **OpenAI**, **DeepSeek**, and **HuggingFace** APIs into **CybORG**, optimizing LLM action-selection workflows to support scalable 500-step adversarial simulations; RL agents achieved $\sim 104\times$ faster action selection (45s vs. 4,705s per episode)

PUBLICATIONS

LLMs are Autonomous Cyber Defense Agents | *IEEE Conference on Artificial Intelligence (CAI 2025)*

- IEEE Conference on Artificial Intelligence (CAI 2025) — arXiv:2505.04843